Deepfake Forensic Analysis Report

AI-assisted forensic report generated from model prediction, Grad-CAM attention, quality analysis, and explainable AI interpretation.

1. Case Information

Analysis ID	111
Filename	example-17.png
Final Prediction	FAKE
Confidence	50.65%
Faces Detected	1
Selected Face	Face #1

2. Model Result Summary

Real Probability	49.35%
Fake Probability	50.65%
Brightness	56.14
Blur Score	149.73
Risk Level	LOW

3. Grad-CAM Heatmap Evidence



Highlighted regions show where the model focused during prediction. These regions represent AI attention zones, not confirmed manipulated areas.

4. Region Attention Analysis

Region	Label	Attention Score

R1	Nose Bridge Region	99.03%
R2	Mouth and Lip Region	95.96%
R3	Left Eye Region	74.91%

5. XAI Textual Explanation

• Model attention was strongest around Nose Bridge Region.

• The marked Grad-CAM regions show where the model focused during prediction.

• These highlighted regions are attention zones, not confirmed manipulated areas.

6. Forensic Explanation

The selected face is classified as FAKE with 50.65% fake probability. The explanation is restricted to face-landmark regions only.

Manipulation Indicators

• The model used the nose bridge as an evidence region (99.03% attention). This region contributed to the fake classification because it relates to central facial alignment and texture continuity.

• The model used the mouth and lip region as an evidence region (95.96% attention). This region contributed to the fake classification because it relates to lip boundary, expression shape, and lower-face texture consistency.

• The model used the eye region as an evidence region (74.91% attention). This region contributed to the fake classification because it relates to eye alignment, reflections, eyelid texture, and local facial consistency.

Quality Observations

• Brightness is within the acceptable analysis range.

• Blur score is acceptable for facial analysis.

Recommendation

Review the highlighted face-landmark regions first. These are model attention zones, not confirmed fake pixels.

7. Audit Trail

Stored Image	/static/uploads\77014793388a47a78844f38f0ae131c2.png
SHA-256	ff2bc02ffc1dec85e65b3a548e0043f58334a3a9a59830200b39a80e86f74cad
Report Type	Automated forensic analysis report

Disclaimer: This report is generated by an automated AI-based forensic system. The result should be reviewed by a qualified digital forensic expert before use in formal investigation or legal proceedings.